

$$3d^9$$

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$$1 \quad \left| 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 3d_{xz}^\alpha \right\rangle$$

Here are the CSFs:

$$\begin{aligned} |1, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha \right\rangle \\ |2, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2 \right\rangle \\ |3, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\ |4, 1/2, 1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\ |5, 1/2, 1/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\ |6, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta \right\rangle \\ |7, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2 \right\rangle \\ |8, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\ |9, 1/2, -1/2\rangle &= \left| 3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \\ |10, 1/2, -1/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 \right\rangle \end{aligned}$$

The Ground State CSFs are 5 10

$$\langle 5, \frac{1}{2}, \frac{1}{2} | L_x | 3, \frac{1}{2}, \frac{1}{2} \rangle = I$$

$$\Delta g_{xx}$$

$$+2\zeta\Delta_{5-3}^{-1}$$

$$\begin{aligned}\langle 5, \tfrac{1}{2}, \tfrac{1}{2} | L_y | 1, \tfrac{1}{2}, \tfrac{1}{2} \rangle &= -I \\ \langle 5, \tfrac{1}{2}, \tfrac{1}{2} | L_y | 2, \tfrac{1}{2}, \tfrac{1}{2} \rangle &= \sqrt{3}I\end{aligned}$$

$$\Delta g_{yy}$$

$$+2\zeta\Delta_{5-1}^{-1}$$

$$+6\zeta\Delta_{5-2}^{-1}$$

$$\langle 5, \tfrac{1}{2}, \tfrac{1}{2} | L_z | 4, \tfrac{1}{2}, \tfrac{1}{2} \rangle = -I$$

$$\Delta g_{zz}$$

$$+2\zeta\Delta_{5-4}^{-1}$$